

Digital Twin as Enabler of Sustainability and Risk Management of Medical Devices

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Abstract. This paper discusses how the implementation of a digital twin approach can assist organizations in attaining two related but non-competitive objectives: promoting transparent risk-based decision-making and demonstrating that sustainability objectives can be accomplished in a more direct and cost-effective way. When developing a digital twin, it is crucial to consider its intended use, anticipated advantages for the organization's personnel and processes, and usage. Creating a digital twin of a medical device, its aggregates, and components has significant downstream impacts, including affecting supplier partnerships and influencing design and manufacturing processes. Through its simulation and training features, it can bolster users' and patients' confidence in using the technology and simplify the medical staff's job. Furthermore, a digital twin can serve as a repository for the knowledge required for treatment, medicine, and monitoring the patient's condition's progress. This paper highlights a digital twin implementation which case-focused or goal-driven, thereby simplifying activities such as collaborating with a supplier. However, it also necessitates the participation of multiple stakeholders and may affect their behavior..

Keywords. Product Risk Management, Configuration Management, Modular Design, Product Lifecycle Management, Transdisciplinary Engineering

Introduction

Sustainable development, the prominent concept applied in many fields today, aims to improve human well-being while preserving the natural systems that support life on Earth. This concept was introduced globally in 1987 by the World Commission on Environment and Development in a report called "Our Common Future" [1]. In 1992, the United Nations Conference on Environment and Development, also known as the "Summit of Rio," endorsed this idea and emphasized the importance of academic institutions in actively addressing environmental and developmental issues. Today, sustainable innovation has become a dominant principle of development worldwide,

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extending beyond its initial impact and deployment in the energy and transportation sectors, as dictated by international conventions [2].

Since the inception of lean innovation principles, which aim to generate value without waste, the framework conditions for companies in the manufacturing industry have undergone a significant transformation. This has occurred amidst an increasingly competitive global environment characterized by heightened volatility, uncertainty, complexity, and ambiguity, collectively referred to as VUCA, thereby presenting significant challenges for modern enterprises [3]. Sustainability requirements have become ubiquitous for companies that cater to end-customers directly and for suppliers operating within the value chain. Although literature reports several sustainability metrics approaches [4], the precise definition of the target values related to sustainability requirements is not always clearly defined. Companies strategically operationalize terms such as "sustainability" and "climate neutrality," and their definitions are in a constant state of flux. Sustainability indicators are case-specific and typically derive from various sources such as strategic considerations within the company, local factors, country-level regulations, and trade agreements, among others. The importance of flexibility in defining these indicators has become increasingly evident for companies' individual success and for achieving overall sustainability targets.

Risk is commonly understood as an identified yet potentially unfavorable outcome, whose likelihood can be approximated [5]. The probability of risk is intrinsically associated with uncertainties at multiple levels, including the uncertainty of objectives not being achieved, the uncertainty of events occurring, and the uncertainty of the consequences resulting from anticipated impacts. This phenomenon arises when a decision has several feasible outcomes, and the possibility of unfavorable outcomes is unclear due to insufficient information or volatile variables within the environment [6].

Risk is an inherent aspect of a company's activities at all levels and can lead to observable degradation in the quality of products and services, consequently diminishing competitiveness [7]. Consequently, risk is frequently subject to internal and external regulations and audits, often carried out by regulatory bodies [8]. However, the description and evaluation of risk can vary significantly, contingent upon the specific business area of application, industry, and domain.

Digital twins (DTs) are virtual replicas of products, processes, or services that encompass a highly detailed representation of their physical counterparts. The goal of DT is to amalgamate the benefits of twinning, simulation, real-time monitoring, analytics, and optimization. DT has been identified as the subsequent breakthrough in digitization, as well as the next wave in simulation. By facilitating testing on a virtual prototype, DT can save costs, time, and resources for prototyping, thereby eliminating the need for developing physical prototypes, without impacting actual operations [9].

In the business environment, marked by an incessant need for innovation, DTs can serve as a tool for innovation and as an innovative idea itself. DTs can be used to achieve stakeholder involvement, adaptation of business models, incorporation of environmental factors, disruptive technologies, digitalization of services and operations, and engagement with customers. These factors offer potential fields for the implementation of DTs [10]. In terms of digital transformation, DTs have become a common tool for enhancing efficiency, reducing costs, and increasing profitability. Advancements in Industry 4.0 technology have been leveraged to implement sustainable practices [2].

The issues raised here arise at one of the world's largest vendors of medical devices, which operates in a highly regulated environment and where any product flaws could harm the patient's health. In this setting, close cooperation with regulators is essential.

In this paper, we will propose how digital twin can enable the sustainability and risk management of medical devices. The outline of the paper is as follows. In section 1, the background of risk management and PLM is presented. In section 2, the outcome of digital twin, especially for medical devices are derived. We discuss how digital twin works in practical, collaboration context in section 3. Discussion of the challenges that still exist is presented in section 4. The paper ends with a brief conclusion and outlook.

1. Background

Innovation and sustainable development go hand in hand. It is increasingly recognised that economic growth and social progress depend on new ideas and technologies, as well as the ability to bring them to fruition. In this context, innovation can play a key role in achieving sustainability. Internal factors such as the product development process, commercialization and dissemination activities to bring new products and services to customers play dominate the sustainability factors of an enterprise. Furthermore recent research suggests that external factors around sustainable supply chain are also pivotal to these goals [11].

According to a study, 11 interconnected functions of Industry 4.0 facilitate sustainable innovation, leading to the development of a model that specifically addresses sustainable innovation and is the roadmap to sustainable innovation capability development [12]: advanced manufacturing competency, green absorptive capacity development, green process innovation capacity, green product innovation capacity, inter-functional collaboration and learning, new product development competency, product life-cycle management capability, sustainable innovation orientation development, sustainable partnership and collaboration, sustainable talent management, and value chain integration.

Studying resilience, innovation, and sustainability concurrently and their interactions has a lot of potential as these concepts are proven to overlap. Innovation is crucial for creating both resilience and sustainability, but mechanisms for achieving and upholding sustainability entail resilience. When examining resilience, sustainability, and innovation all at once, multi-scale and multi-stakeholder methodologies should be used [13].

Although there are numerous potential risks, using classification, prioritization, and assigning each risk to a certain process in order to determine which ones require urgent attention and how to spend limited resources. The risk assessment establishes the likelihood and effects of each risk in readiness for a subsequent scoring process [14].

To develop a clear method that can be applied in Product Lifecycle Management (PLM), systematic approaches for analyzing the possibility and implications of hazards associated to products are required, even though intuitive approaches for doing so are standard practices for experienced engineers [15].

Systemic risk, coordination, communication, labor relations, outsourcing, and offshore all had a substantial impact on managing risk. Geographical dispersion, stakeholder relations, and cultural diversity are further topics that merit discussion [16]. Some sources provides a general risk management method that takes into account things like budget, objectives, requirements, supplier, legal, and regulatory considerations [17]. Additionally, we shall always consider collaboration as a potential risk factor [18] given the way modern companies operate.

The two inherent capabilities of digital twins—the ability to integrate sizable amounts of static, real-time, structured, and unstructured data, as well as the ability to weave these data with more advanced data processing techniques like artificial intelligence or high-performance computing to put simulation, control, and self-improvement—are what are driving the industry's increased interest in them as a mean to solve complex emergent problems [19].

Digital twins, which we understand as a virtual model of physical products, present unique challenges during each stage of the product's lifecycle, including definition, design, manufacturing or building, and servicing. Each digital twin instance differs based on a unique perspective and set of requirements and the goals it is meant to achieve. Focusing on the various application fields and their associated technologies, an advanced and current picture of the state-of-the-art is presented, taking into account the key characteristics and difficulties of the scientific research on digital twins that has already been conducted [20].

Regarding the discrepancy in objectives, criteria for the existing approaches, particularly the evaluation of indicators, must be carefully chosen. The procedure must take into account the trade-off between the restrictions mentioned during the design stage and, the quality and range of the resulting measures and indicators [4]. The method must be sensitive enough to cover an indicator's full range.

The "RACER criteria" which is often utilized to assess electronic systems, was chosen to establish criteria for evaluating the creation of digital twins for medical devices. However, it was modified to address the specific challenges and goals related to this field. The criteria are highlighted in Table 1 [4]:

Table 1. RACER criteria for evaluation of electronic systems.

Criterion	Meaning
Robust	Reproducible data; Comparable and applicable to further products or new generations of products
Accepted	Accepted by stakeholder (in particular: designers) by means of applicability on product level; Applicability to medical devices..
Credible	Easy to evaluate, interpret and justify.
Easy	Minimum input data required; Data availability at design stage; Low calculation time.
Relevant	Quantitative data; Based on chemical and physical characteristics; Linked to environmental impacts.

The available indicators can be sorted into various dimensions. During the evaluation process, indicators were identified based on the RACER criteria and gathered in an Excel spreadsheet. In order to organize the indicators and simplify the complexity, they were classified into a two-dimensional matrix.

The first dimension of the matrix employs a lifecycle assessment (LCA) approach, which utilizes several use cases to structure the stages of a medical device's lifecycle. Specifically, we selected four use cases that are highly relevant in daily operations. To balance the importance of the stages with data availability, we conducted interviews with stakeholders.

The second dimension of the matrix addresses the objectives that can be achieved by generating a digital twin. For this paper, we focused on two objectives that are unique to medical devices: sustainability and risk management. Because these objectives are mutually dependent in certain cases, we evaluated them using three characteristics, which are highlighted in Table 2.

2. Benefits of Digital Twin

A digital twin has three main advantages. First, virtual models powered by real-time data have the ability to replicate the real world and infrastructure with a high degree of authenticity. As a result, they may be designed to allow a variety of interactions between machines or between humans and machines, including immersive technologies. Second, a digital twin enables the user to obtain a thorough description of the entire environment and a greater comprehension of the physical entities by integrating, merging, and analyzing actual and simulated data. Lastly, digital models offer succinct, intuitive, and simple-to-use platforms like front-end software and mobile apps that people in various fields could use.

As shown in Table 2, we have determined the associated traits for each combination of four exemplary use cases and two objectives in three tiers in order to categorize these advantages. Of course, this list of qualities is not complete, and more use cases may need evaluation.

Table 2. Evaluation matrix for four use cases.

	Sustainability	Risk management
Supplier selection	Early, collaborative performance review	Supplier and manufacturer sharing risk
	Reliable component quality	Controlled component risk uncertainty
	Reduced scrap material due to quality issues	Risk management dependability of supplier output
Lifespan prediction	Higher product reliability	Decreased chance of malfunction
	Reduced warranty coverage for recalled products	Improved management of service information
	Updating products more quickly to reflect changes in usage	High level of product usage and handling knowledge
Auditability	Quicker and more trustworthy auditing data available	Simplified certification
	Digital presentation of changing product	Better risk quantification as a result of modifications.
	Build confidence	Transparency in the assessment of risk
Ecological footprint	Exact prediction of metrics	Metric success motivates additional risk analyses
	Operational case prediction	A higher standard of customer service and updated security threats
	More satisfied customers and fewer improperly disposed of devices	Evaluation of product disposal risk

2.1. Supplier selection

Suppliers are given a more important role in product development while vertical integration continues to diminish [21]. An accurate assessment of performance in relation to the components they offer is one of the most crucial criteria for choosing a supplier. Decision frameworks for evaluating sustainable, innovative suppliers are already available [22].

In terms of choosing suppliers, the digital twin reduces the quantity and scope of tests while improving sustainability in the crucial area of performance evaluation of a given component. With these outcomes, it will be possible to decrease the number of suppliers, raise the lot size, and decrease the purchase price.

The digital twin helps risk management by enabling early risk detection in supplier collaboration. The identification of crucial phases is then made easier by the digital twin. The risk distribution between the vendor and its suppliers can be optimized with the help of this information.

2.2. Functional assessment: lifespan prediction

However, it is unclear what an extensive high-fidelity DT might actually look like, which would optimize services while minimizing cost and technical implementation difficulties, and mimic every feature of the physical system [23].

The lifespan may be exactly anticipated with such a digital twin, increasing reliability. Accurate behavior prediction results in lower warranty costs. A precise prediction boosts consumer satisfaction overall and adds to sustainability as a whole.

The early detection of problems that could harm the device's performance and, as a result, the patient's health, is made easier by the digital twin. Customer service can be better managed as a result. Finally, there are fewer unforeseen situations that the risk management must handle.

2.3. Auditability

Since it is a digital replica of the medical device, the digital twin is naturally suitable for audit. It offers trustworthy evidence of the effectiveness of the product, and the digital twin contains detailed records of the modifications that were made, the justification for them, and a brief assessment of the situation before and after the change [24].

As a result, greater transparency and less labor-intensive information production improve the auditability's sustainability component. As a result, costs are reduced. Finally, it increases the auditor's confidence.

The digital twin helps risk management by foreseeing problems that are important to auditors. Risk assessments and quantification can then be better conveyed during the audit. The digital twin allows for an immediate response to the audit requirements.

2.4. Ecological footprint

The ecological footprint is increasingly playing a crucial part in businesses when it comes to product performance, in addition to the traditional evaluation aspects of cost, time, and quality. Customers are increasingly seeking ecologically friendly items in addition to legislative obligations [25].

The digital twin offers precise metrics prediction for the ecological footprint to increase sustainability. Exact operating scenario calculations can further improve sustainability. With the help of these numbers, a precise waste forecast may be made.

The digital twin aids in risk management by giving the data required for a smooth certification process.

The implementation of DTs can contribute significantly to the sustainability goals and goals in general of a company throughout the lifecycle of the product. For example, by reducing material waste and energy consumption during the product development;

maximizing energy consumption reducing the amount of battery usage; reducing raw material consumption during manufacturing; optimizing supply chain and reducing the environmental impact of product maintenance and disposal.

3. Digital Twin in practice

A case study was selected to showcase the practical advantages of digital twin technology in engineering collaboration with suppliers, specifically for sustainability and risk management purposes [26]. The case focused on a medical device's vital component supplied by external manufacturers which significantly affected its proper functioning. This example demonstrated how the digital twin can be used effectively in supplier selection, lifespan prediction, auditability, and ecological impact assessment.

A wearable medical device comprises essential modules developed by various facilities and suppliers. Ensuring the product quality meets expectations is critical to avoid health risk for patients or users. The power module, which includes an interchangeable battery meeting specific requirements such as a three-week typical use supply, holds paramount importance in this setup. Several factors comprise the model for the battery output: Use pattern and frequency; environmental factors, such as humidity and temperature; features triggered during use.

The R&D department makes significant investments in battery tests in order to gather data to ensure that the device is usable for the requisite period. With the help of this data, design choices are made to match typical, anticipated user profiles. Standard profiles fit the criteria and specifications' edge cases. The tests are based on batteries that are readily available on the market because the producer purchases batteries there. Although these batteries' official specifications are comparable, they perform very differently.

It is significant to remember that the project creating this product is doing so in an agile setting. The test's scope and breadth must therefore change to fit the current sprints. It was decided that comprehensive testing was not practicable since new battery load scenarios might emerge throughout the development process. The R&D department made the decision to develop a DT that reflects the actual design information of the medical device in order to support the short-cycled simulations required for agile releases (marked with an arrow in Figure 1).

The R&D department collaborates with a provider that has substantial experience in the field of battery simulators, particularly in other industries like transportation, to ensure that the tests are thorough enough to produce accurate and objective results (Figure 2). utilizing as a starting point the criteria and specifications that the R&D division transparently controls in an application lifecycle management system (ALM). The availability and costing criteria describe the batteries on the market that will be taken into consideration when creating the DT. These requirements are utilized to construct a variety of usage profiles.

The provider then searches the market for batteries that substantially satisfy the specified specifications to create a short list. The short-listed batteries were afterwards purchased from the market and put through testing in a lab. A characterization of the batteries is the outcome. In a subsequent phase, the characterization is verified using physical testing on various medical equipment conducted under various usage and environmental conditions. The risks related to the chosen batteries and the test results will also be brought up at this point and logged. The battery DT's mathematical foundation is derived from the validated results. An impact study will be performed

during the development process and every design modification to see if the battery is affected, while taking into account any hazards that need to be updated or reduced. Only when the modification is supported by simulation results will an update of the design lead to an update of the DT simulation. A modification in the design could have an impact on the typical usage profiles and, consequently, on how much battery is used.

The DT simulation's findings can support the model or call for changes to it. The DT simulates the short-listed brands since it is possible that an updated standard profile might be better satisfied with a particular brand. The outcomes of the simulations are all kept in a dedicated data repository. For later traceability, the information is linked in the PLM system to materials, batches, versions, studies, etc. This includes how well the device function in relation to the risks they pose. The new version of the battery item is mentioned in the documentation, and the item and consequently the associated information can only be formally released when the predefined release requirements are satisfied. Which battery in which version will be a component of the released product will depend on the item. The following iteration of the physical device may then be constructed using released goods when they have been purchased. A dashboard makes the compilation of all this data accessible for quick and clear decision-making. Every modification to the requirements' design will start the aforementioned procedure.

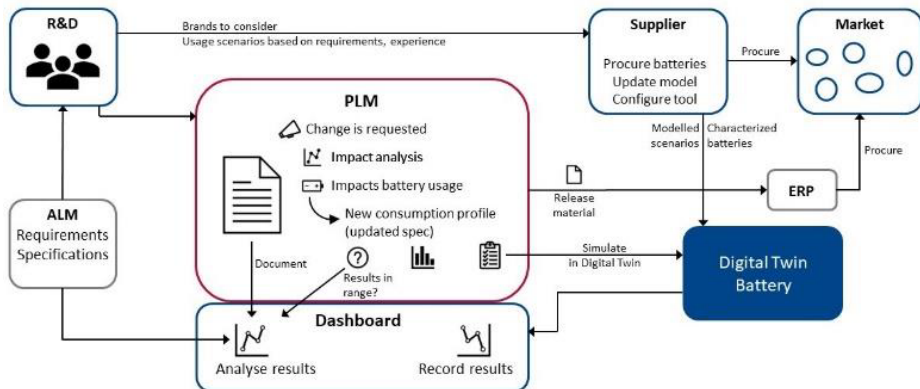


Figure 1. Use case Battery Digital Twin.

The development process is the main objective of this model. The next stage is to expand the model to collect real-time data from the devices utilizing internet-of-things. With the benefits to be able to detect issues as they occur and predict the need for maintenance. The same information is very valuable to the product development process. Safeguarding patient information, keeping it secure and anonymous and sorting are associated challenges.

4. Discussion

The intended use and use of the digital twin, as well as the anticipated benefits for the organization's people and processes, must all be taken into account when creating it [27]. In this view, it is important to consider the following downstream effects of a digital twin of a medical device, including the device as a whole, as well as each component: It might have an effect on working with the suppliers, as well as the design and production processes. By its simulation and training features, it can encourage an increase in

user/patient confidence and make the job of the medical team easier. Also, the digital twin can serve as a repository for the knowledge required for diagnosis, therapy, and analysis of the disease progression of the patient [28].

The introduction of DTs pose technical and organizational challenges, thus new technologies and processes must be adopted. This is valid also for suppliers, especially if they are partnering with several manufacturers as many of these new processes and technologies are not standardized. For end-users they may be confronted with the privacy-convenience trade-off dilemma. Since the likelihood of unintended data exchange increases, proper privacy protection measures must be offered. Particularly, the definition of urgency and medical ethics and law may be significant factors [27].

So, it is reasonable to anticipate that the digital twin will eventually serve as both a requirement for each medical device's certification and a way for regulatory bodies to undertake audits more quickly and cheaply. This might result in the creation of a "certification twin" that might be applied to the evaluation and certification of suppliers, as described in section 3.

5. Conclusions and outlook

Using digital technologies, it is possible to improve the effectiveness and resourcefulness of processes and goods. Digital platforms can also be utilized to connect industries and sectors and take advantage of synergistic effects amongst them. Moreover, digital technologies provide the chance to connect current supply and recycling channels into circular value chains.

This will allow for the application of digital technologies in product development to create sustainable products. The employed instruments ought to offer a clear foundation for quicker and better judgments. With the use of sustainable products for the creation of circular value, this should facilitate ecological transformation.

A digital twin is a suitable method for offering an architectural foundation for future medical device development in terms of the device's effectiveness, safety, hazards, and sustainability. Yet, sustainability is typically not defined in such specific terms and is particularly variable, necessitating the use of suitable models and methodologies.

Gain the sustainability and risk management to the primary objectives with the mission to ensure the patient's health and well-being by providing enough therapy and care on a high level of quality and sustainability. The application of engineering and IT solutions to augment or replace specific human functions opens a quickly expanding path for further growth by employing methodologies of system science and systems engineering that are employed in other industries.

The possibility for additional advancements is significantly increased by an integrated application that also includes the procedure. The use of digital twins appears to be a logical outcome of customized medicine supported by artificial intelligence. A transdisciplinary strategy that takes into account all the relevant factors—from technical expertise to personnel management to economics and innovation—would encourage the myriad societal requirements and serve as the foundation for ongoing adaptation.

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